**National University of Computer and Emerging Sciences,**
**Lahore Campus**

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| | **Course Name:** | **Computer Networks** | **Course Code:** | **CS 3001** |
| | **Program:** | **BS (Software Engineering)** | **Semester:** | **Spring 2026** |
| | **Duration:** | **20 minutes** | **Total Marks:** | **15** |
| | **Paper Date:** | **16-April-2026** | **Section** | **6A , 6B** |
| | **Exam Type:** | **Quiz 3 - Chapter 3** | **Page(s):** | **2** |

**Student Name** **Roll No.** **Section:**

**Q1.** Host A establishes a TCP connection with Host B using the three-way handshake. Host A selects an initial sequence number (ISN) of 1200, while Host B selects an ISN of 4800. Immediately after the connection is established, Host A requests to send 5000 bytes of data to Host B. Given that the Maximum Segment Size (MSS) is 1000 bytes: **[3+1+1 = 5 Marks] [CLO 2]**

(a) Draw the 3-Way Handshake diagram, labeling all SYN bits, ACK bits, Seq numbers, and Ack numbers.

(b) Show the first two data segments sent by Host A. What are their Sequence Numbers?

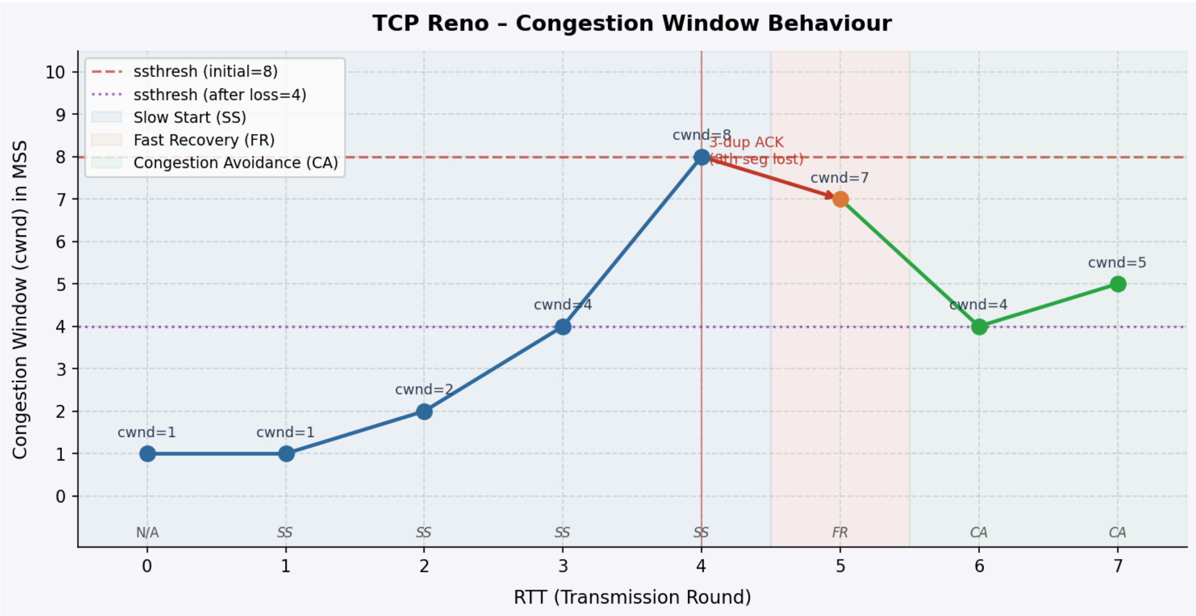
(c) If the second data segment is received but the first is delayed, what Ack number will Host B send in response to the second segment?

**Q2.** Consider a TCP Reno connection as illustrated by the diagram below. Note the following:

- 1 MSS = 100 bytes
- At RTT = 0, the TCP connection has been established but no segment has been sent. Initial values are given in the table below.
- The sender starts transmission (first segment, seq. no. = 0) at RTT = 1.
- The 8th segment sent by the sender is lost.
- Out-of-sequence received segments are discarded (not buffered).
- SS = Slow Start, CA = Congestion Avoidance, FR = Fast Recovery, N/A = Not Applicable.
- Loss detected via 3 Duplicate ACKs.

(a) Fill in all missing entries in the TCP Sender Table (Table 1) below. Some entries are pre-filled for guidance.

(b) What is the sequence number of the segment that was lost? ____________________




| TXNR(RTT) | rwnd (MSS) | cwnd (MSS) | ssthresh(MSS) | RecvACKNo. | Seq No(s) Transmitted | Timer (ms) | NoLossOrTimeoutOr3dupACK | TCPMode(SS/CA/FR) |
|---|---|---|---|---|---|---|---|---|
| 0 | 8 | 1 | 8 | N/A | N/A | 600 | No Loss | N/A |
| 1 | 8 | | | N/A | | 600 | No Loss | SS |
| 2 | 8 | | | | | 600 | | |
| 3 | 8 | 4 | | | | 600 | No Loss | |
| 4 | 8 | | 8 | 700 | | 600 | | |
| 5 | 8 | | | | 700 | 600 | | |
| 6 | 8 | | | | | 600 | No Loss | |
| 7 | 8 | 5 | | | | 600 | No Loss | CA |